The Lottery Ticket Hypothesis

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The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks - MIT CSAIL
Yannic Kilcher - <https://www.youtube.com/watch?v=ZVVnvZdUMUk>

THE LOTTERY TICKET HYPOTHESIS: FINDING SPARSE, TRAINABLE NEURAL NETWORKS

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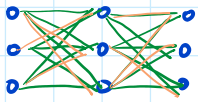
Sort of an empirical paper into what makes NN train successfully
And it comes out of the literature of pruning

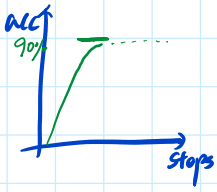
ABSTRACT

Neural network pruning techniques can reduce the parameter counts of trained networks by over 90%, decreasing storage requirements and improving computational performance of inference without compromising accuracy. However, contemporary experience is that the sparse architectures produced by pruning are difficult to train from the start, which would similarly improve training performance.

성능을 유지하면서 model을 작게
가져갈 수 있을까??
⇒ pruning의 배경

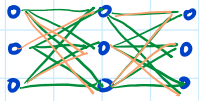
1. Full Network train





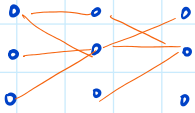
큰 파라미터의 모델이 학습하는 것은 좋은 intialization의 조합을 찾도록 하는 것으로 여겨질 수 있음

2. Prune
- select some connections
  많은 방법들 有.
- quantization.
  distillation 연구.



transfer over the weights of full trained network as initiallizations of subnetwork

3. Re train
2단계에서 Pruning 된 subnetwork 를
"Under one condition" 하에서 재학습한다



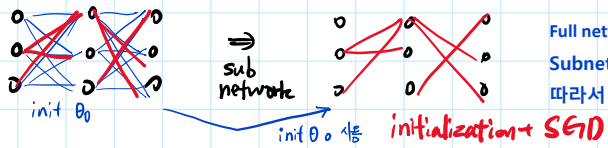
Good initialization으로 시작해야 하기 때문에 weights를 가져온다.

이전에 parameter가 가지고 있던 responsibility를 다른 weight가 분담하면서 더욱 훌륭한

In this paper, we show that there consistently exist smaller subnetworks that train from the start and learn at least as fast as their larger counterparts while reaching similar test accuracy. Solid lines in Figure 1 show networks that we find. Based on these results, we state *the lottery ticket hypothesis*.

The Lottery Ticket Hypothesis. *A randomly-initialized, dense neural network contains a subnetwork that is initialized such that—when trained in isolation—it can match the test accuracy of the original network after training for at most the same number of iterations.*

More formally, consider a dense feed-forward neural network $f(x;\theta)$ with initial parameters $\theta = \theta_0 \sim \mathcal{D}_\theta$. When optimizing with stochastic gradient descent (SGD) on a training set, f reaches minimum validation loss l at iteration j with test accuracy a . In addition, consider training $f(x; m \odot \theta)$ with a mask $m \in \{0,1\}^{|\theta|}$ on its parameters such that its initialization is $m \odot \theta_0$. When optimizing with SGD on the same training set (with m fixed), f reaches minimum validation loss l' at iteration j' with test accuracy a'. The lottery ticket hypothesis predicts that $\exists\, m$ for which $j' \leq j$ (*commensurate training time*), $a' \geq a$ (*commensurate accuracy*), and $\|m\|_0 \ll |\theta|$ (*fewer parameters*).



Full network로 하여금 성능이 좋은 intialization의 Subnetwork 조합을 찾아내도록 함.
따라서 초기의 많은 파라미터가 그 수많은 조합들의 set같은 느낌

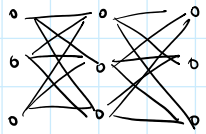
Identifying winning tickets. We identify a winning ticket by training a network and pruning its smallest-magnitude weights. The remaining, unpruned connections constitute the architecture of the winning ticket. Unique to our work, each unpruned connection's value is then reset to its initialization from original network *before* it was trained. This forms our central experiment:

1. Randomly initialize a neural network $f(x;\theta_0)$ (where $\theta_0 \sim \mathcal{D}_\theta$). ⇒ 어느 params를 제거해야할 지 알기 위해 full network를 학습시켜야..
2. Train the network for j iterations, arriving at parameters θ_j. trained params
3. Prune $p\%$ of the parameters in θ_j, creating a mask m.
4. Reset the ~~remaining~~ parameters to their values in θ_0, creating the winning ticket $f(x; m \odot \theta_0)$. same with beginning params. masked version.

As described, this pruning approach is *one-shot*: the network is trained once, $p\%$ of weights are pruned, and the surviving weights are reset. However, in this paper, we focus on *iterative pruning*, which repeatedly trains, prunes, and resets the network over n rounds; each round prunes $p^{\frac{1}{n}}\%$ of the weights that survive the previous round. Our results show that iterative pruning finds winning tickets that match the accuracy of the original network at smaller sizes than does one-shot pruning.

iterative pruning이 one-shot pruning에 비해 더 작은 size로 original network의 Acc를 match.

e.g.



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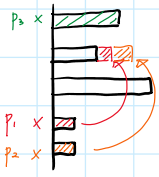
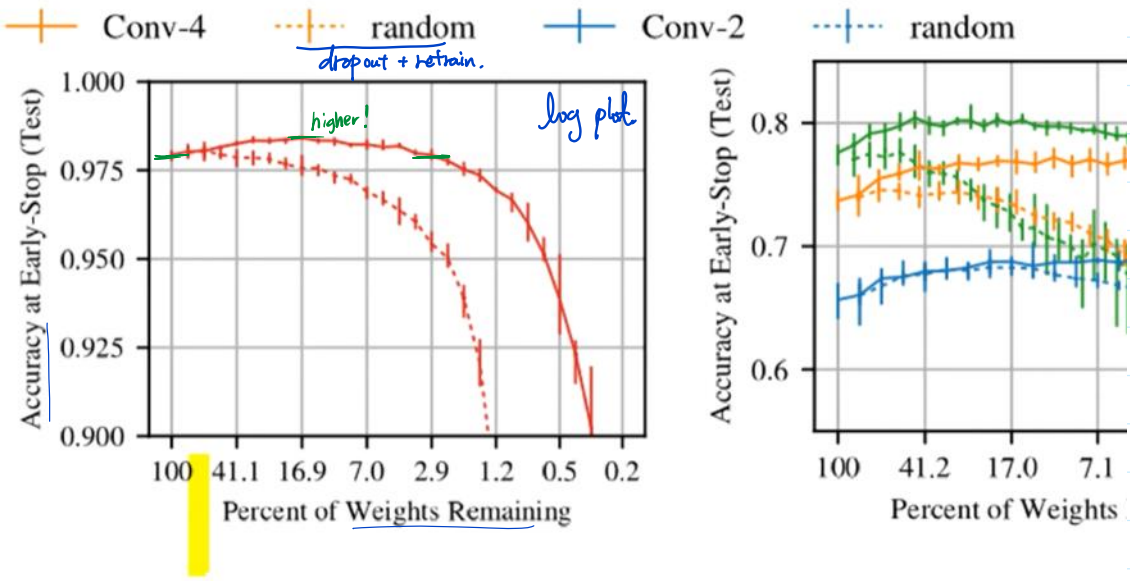
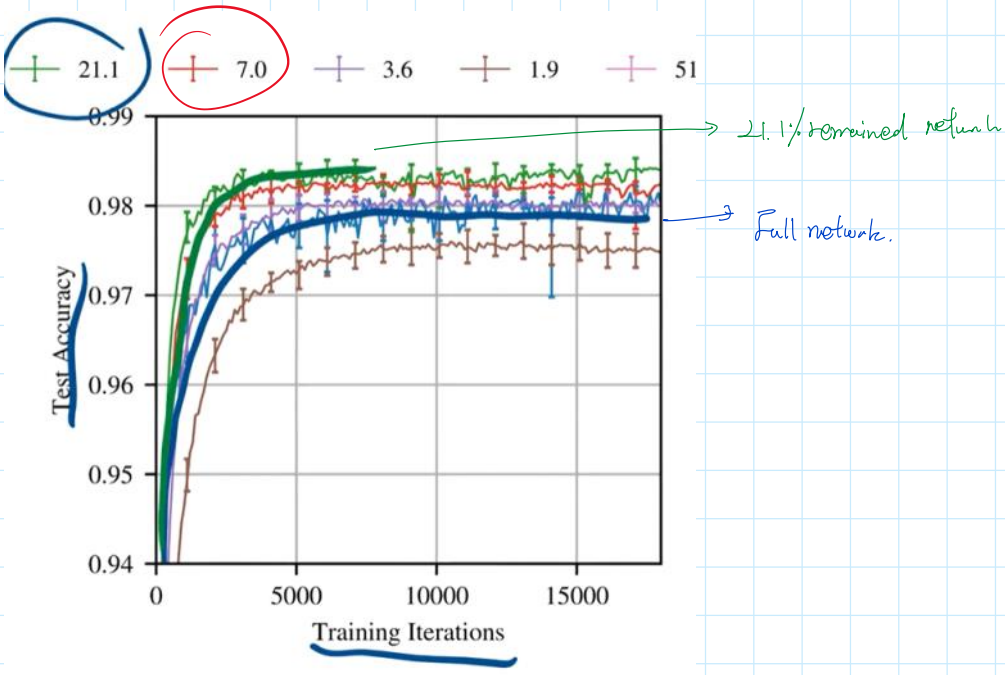


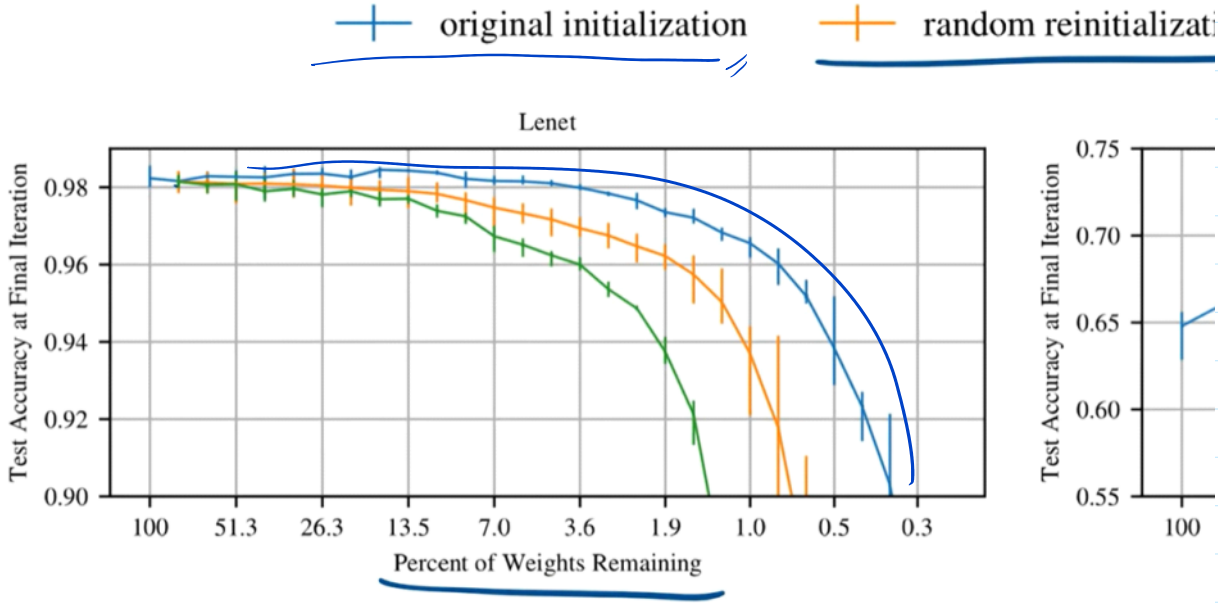
이런 식으로 iterative pruning 진행.
이전 params가 가지고 있던 responsibility를 다른 params가 분담하면서 더욱 훌륭한 params 들만 남도록 pruning.

parameter 들만 남게 되는 것.

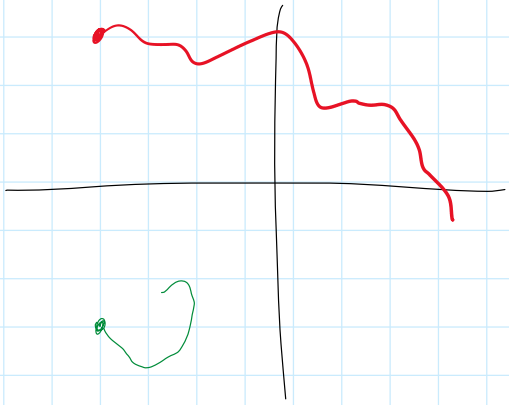


plot sorry that that deals with percent





random reinitialization이 아니라, original initialization을 가져가는 것이 더욱 성능에 좋음.



$\theta_0 \rightarrow \theta_f$

$\theta_1 \rightarrow \theta_{f'}$

초기의 intilization을 그대로 가져가면서 pruning을 해가는 것이 Global minima를 찾는 데에 더욱 도움이 된다는 것을 보여주는 듯